

OFFLINE (development)

Collect + label

25 recordings
15 pos / 10 neg

Feature + window

resample, 60 ch,
200-sample windows

Train TCN

multi-task loss;
LORO evaluation

Calibrate

temperature +
operating point

Export

ONNX model +
norm. & calib. stats

deploy: ONNX + normalization + calibration statistics

ONLINE (on-robot, ROS 2)

Unitree Go2

/lowstate stream
BEST_EFFORT QoS

entanglement_detector node

ring buffer → features → normalize → ONNX
→ calibrate → debounce → stationarity gate

/entanglement_state

flag, confidence,
per-leg prob + severity